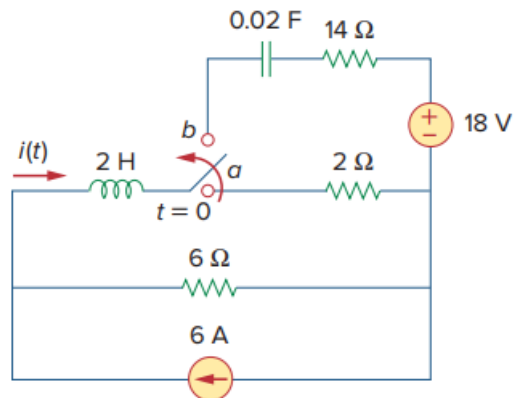


Problem

The switch in the circuit of Fig. 8.88 is moved from position a to b at $t = 0$. Assume that the voltage across the capacitor is equal to zero at $t = 0$ and that the switch is a make before break switch. Determine $i(t)$ for all $t > 0$.

Figure 8.88:



Step-by-step solution

Step 1 of 9

Refer to the Figure 8.88 in the textbook, .

At $t = 0$, the switch is moved from position a to b .

The switch is at position a for $t < 0$.

At $t = 0^-$, the circuit is at steady state. The inductor acts as short circuit and capacitor acts as open circuit.

Redraw the circuit at $t = 0^-$ as shown in Figure 1.

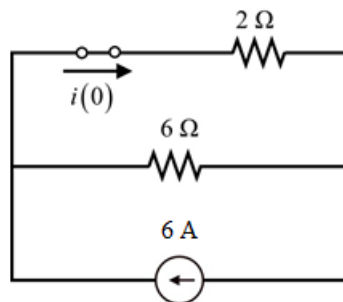


Figure 1

Step 2 of 9

Apply current division rule to calculate the initial current through the inductor.

$$\begin{aligned} i(0) &= \left(\frac{6}{6+2} \right) (6) \\ &= \left(\frac{6}{8} \right) (6) \\ &= 4.5 \text{ A} \end{aligned}$$

The initial voltage across the capacitor is,

$$v(0) = 0 \text{ V}$$

Step 3 of 9

For $t > 0$ the switch moves from the position a to b.

Redraw the circuit for $t > 0$ as shown in Figure 2.

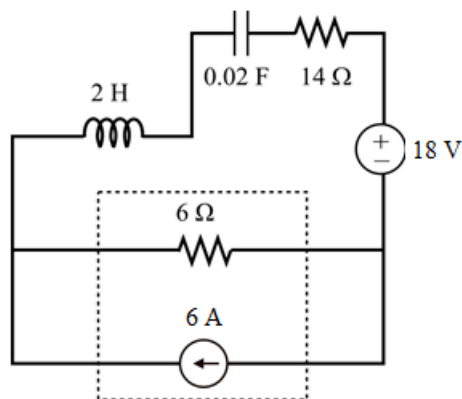


Figure 2

Step 4 of 9

Apply source transformation shown in dotted line to form series RLC circuit.

$$\begin{aligned} V &= IR \\ &= (6)(6) \text{ Convert the current source of 6 A into voltage source.} \\ &= 36 \text{ V} \end{aligned}$$

Draw the modified circuit diagram.

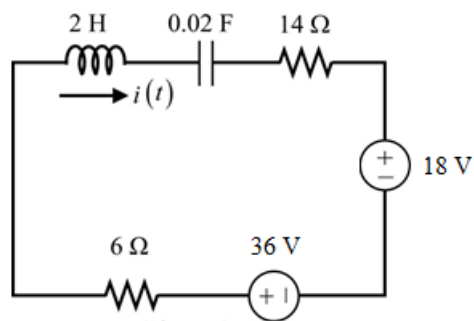


Figure 3

Step 5 of 9

The voltage sources in series can be added and add the resistors in series.

Draw the modified circuit diagram as shown in Figure 4.

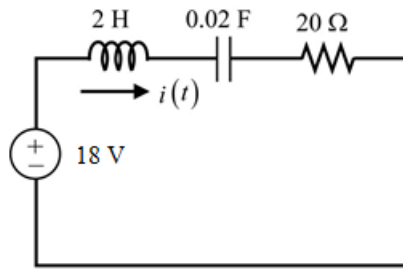


Figure 4

Step 6 of 9

The circuit in figure 4 represents a series *RLC* circuit.

Calculate the value of Neper frequency, α .

$$\begin{aligned}\alpha &= \frac{R}{2L} \\ &= \frac{20}{4} \\ &= 5\text{s}^{-1}\end{aligned}$$

Calculate the value of natural frequency, ω_0 .

$$\begin{aligned}\omega_0 &= \frac{1}{\sqrt{LC}} \\ &= \frac{1}{\sqrt{(2)(0.02)}} \\ &= 5\text{ rad/s}\end{aligned}$$

Since $\alpha = \omega_0$, the system is critically-damped and the solution is of the form;

$$v(t) = V_s + (A_1 + A_2 t)e^{-\alpha t} \text{ V} \dots\dots (1)$$

Where,

A_1 and A_2 are two arbitrary constants.

V_s is the final value of the voltage across the capacitor.

Step 7 of 9

Here V_s is the final value of the voltage across the capacitor.

Calculate the steady state voltage across the capacitor.

At $t = \infty$, the capacitor acts as open circuit.

$$V_s = 18 \text{ V}.$$

The expression for the output voltage is modified as,

$$v(t) = 18 + (A_1 + A_2 t) e^{-5t} \text{ V} \dots\dots (2)$$

Calculate the value of constants A_1 and A_2 using the initial values $v_c(0^-) = 0 \text{ V}$ and

$$i_L(0^-) = 4.5 \text{ A}.$$

Now at $t = 0$, the capacitor voltage is,

$$v(0) = 18 + (A_1 + 0) e^0 \quad [v(0) = 0 \text{ V}]$$

$$18 + A_1 = 0$$

$$A_1 = -18$$

The current through the capacitor at $t = 0$ is,

$$i(0) = C \frac{dv(0)}{dt}$$

$$4.5 = (0.02) \frac{dv(0)}{dt}$$

$$\frac{dv(0)}{dt} = 225 \text{ V/s}$$

Step 8 of 9

Differentiate the equation (2) and substitute 0 for t , -18 for A_1 .

$$\frac{dv(0)}{dt} = -5(A_1 + A_2 \times t) e^{-5t} + A_2 e^{-5t}$$

$$225 = -5(-18) + A_2$$

$$A_2 = 135$$

Substitute -18 for A_1 and 135 A_2 in equation (2).

$$v(t) = 18 + (135t - 18) e^{-5t} \text{ V}$$

The expression for the capacitor voltage is,

$$v(t) = 18 + (135t - 18) e^{-5t} \text{ V}$$

Step 9 of 9

Calculate the current through the circuit.

$$\begin{aligned}i(t) &= C \frac{dv}{dt} \\&= (0.02) \frac{d}{dt} [18 + (135t - 18)e^{-5t}] \\&= (0.02) [0 + (135t - 18)(-5)e^{-5t} + (135)e^{-5t}] \\&= (0.02) [-675te^{-5t} + 90e^{-5t} + 135] \\i(t) &= 0.02(90 - 540t)e^{-5t} \quad \text{Therefore, the current } i(t) \text{ is } \boxed{(1.8 - 10.8t)e^{-5t} \text{ A}}. \\&= (1.8 - 10.8t)e^{-5t} \text{ A}\end{aligned}$$

[Comments \(3\)](#)



Anonymous

The dv/dt should be $(-675te^{-5t} + 225e^{-5t})$, not the simplification that is there.



Anonymous

So the final answer is $(4.5 - 13.5t)e^{-5t} \text{ A}$



Anonymous

Incorrect answer, step 9, line 4, e^{-5t} shouldn't be dropped from 135